NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY
Faculty of Computing and Informatics
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Department of Software Engineering

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Course: Artificial Intelligence—ARI711S

1. Academic Honesty and Integrity

All staff and students of the Namibia University of Science and Technology (NUST), upon signing their employment contracts and registration forms, commit themselves to abide by the policies and rules of the institution. The core activity of NUST is learning, and in this respect, academic honesty and integrity are very important to ensure that learning is valid, reliable, and credible.

NUST, therefore, does not condone any form of academic dishonesty, including plagiarism and cheating on tests and assessments, amongst other such practices. NUST requires students to always do their assignments and to produce their academic work unless given a group assignment.

Academic dishonesty includes, but is not limited to:

- Using someone else's ideas, words, works, or inventions as if it is your work.
- Using the direct words of someone else without quotation marks, even if it is referenced.
- Copying from writings (books, articles, webpages, other students' assignments, etc.), published or unpublished, without referencing.
- Syndication of a piece of work, all or part of an assignment, by a group of students, unless the assignment was a legitimate group assignment.
- The borrowing and use of another person's assignment, with or without their knowledge or permission.
- Infringing copyright, including documents, copied or cut and pasted from the internet.
- Asking someone else to prepare an assignment for you or to write or sit an assessment for you, whether this is against payment or not.
- Re-submitting work done already for another course or programme as new work, so-called self-plagiarism.
- Bringing notes into an examination or test venue, regardless of whether the notes were used to copy or not.
- Receiving any outside assistance in any form or shape during an examination or test.

All forms of academic dishonesty are viewed as misconduct under NUST Student Rules and Regulations. Students who make themselves guilty of academic dishonesty will be brought before a Disciplinary Committee and may be suspended from studying for a certain time or may be expelled. All students who are found guilty of academic dishonesty shall have an appropriate endorsement on their academic record, which will never be erased.

2. Course Information

Course Code and Title: Artificial Intelligence - ARI711S

Department: Software Engineering

Programme: Bachelor of Computer Science (Software Development) [07BCMS]

Notational/Contact Hours: 120 hours

- Contact: 45 hours
- Directed Self-learning and Self-directed Learning: 45 hours
- Assessment: 15 hours
- Supervised tutorial: 10

Mode of Delivery: Face-to-Face

NQF Level and Credit: NQF level 7, 12 credits

Semester Offered: Semester 5

Pre-Requisites: Programming 1 and Applied Statistics for Computing & Informatics

Course Objectives: The course aims to provide students with the foundational principles that drive the applications and practice of implementing AI-based systems.

3. Student Learning

Specific Learning Outcomes:

Upon completion of the course, students will, through assessment activities, show evidence of their ability to:

- Create suitable models for effective and efficient processing using AI techniques to map problem domains into suitable models for effective and efficient processing.
- Evaluate the efficiency and applicability of different AI concepts, models, and algorithms in a specific problem domain.
- Implement AI algorithms.

Comprehensive Learning Outcome

- Evaluate the relevance and applicability of different artificial intelligence concepts, models, techniques, and algorithms in specific problem domains.

4. Course Delivery Methods

- Lectures and demonstrations: Example-driven sessions to clarify concepts.
- Discussion forums: To encourage student participation and collaboration.
- Practical Exercises: Hands-on application on implementing AI algorithms with Python
- Self-Guided Learning: Out-of-class study and use of e-learning resource

5. Assessment and Evaluation

- **Assessment Structure**
  - **Test:** The test will contribute 25% to the CA.
  - **Assignment:** The Assignment will contribute 25 % to the CA.
  - **Group Project:** The group project will contribute 25% to the CA.
  - **Quizzes:** Two quizzes that will contribute 25% to the assessment.

- **Final Mark Requirement**: Students must obtain at least 40% continuous assessment (CA) to write the examination, while a sub-minimum of 40% is required in the examination. The semester mark and examination mark shall be used jointly to determine the final mark in the ratio of 40% (semester mark/CA) to 60% (examination mark). A minimum final mark of 50% is required to pass the course.

- **Supplementary Assessments**: Available for students under specific conditions, with a maximum final mark capped at 40%.
  - The lecturer has the discretion to determine the makeup or supplementary assessments in accordance with the assessment guidelines.

6. Plagiarism Policy

- **Assessment Submission:**
  - The assignment and group project **MUST** be submitted on eLearning to ensure they go through the Turnitin software for plagiarism and similarity checks.
  - The assignment and group report **MUST** be typed and submitted in PDF format.
- **Similarity Thresholds**:
  - **0–20%:** No deduction.
  - **20–40%:** 10% deduction.
  - **40–60%:** 25% deduction.
  - **60–100%:** Automatic failure and disciplinary action.

7. Student Support and Resources

- **Institutional:** NUST Library; Department of Student Services; Centre for Teaching and Learning.
- **Faculty:** Computer laboratories; Moodle repository.
- **Course-Specific:** Course outline, electronic resources, and lecturer consultations.

8. Course Schedule

Week	Dates	Theory (Topics and Sub-Topics)	Activities
1 and 2	09/02/2026 TO 20/02/2026	Course Introduction a) Course Outline b) Software c) Learning platforms	Set up the tone for the course and outline expectations based on the learning outcomes
3, 4 and 5	23/02/2026 TO 06/03/2026	Part 1: What is AI? ○ Introduction to AI ○ Agent Architecture ○ Introduction to Optimization	Tutorials/ Lab
6, 7 and 9	09/03/2026 TO 27/03/2026	Part 2: AI Search Algorithms ○ Uninformed Search (DFS, BFS, UCS, Iterated Deepening Search) ○ Heuristics and Informed Search ○ Local Search (Hill Climbing, Simulated Annealing) ○ Genetic and Evolutionary Algorithms ○ Constraint Satisfaction	Practicals / Lab
10, 11, 12 and 13	30/03/2026 TO 23/04/2026	Part 3: Intro to Machine Learning ○ Information and Data Representation (Tabular, Graph, image, video) ○ Feature Selection ○ Supervised Learning (regression, classification, regularization) ○ Unsupervised learning (clustering, Dimensionality Reduction, etc.) ○ Artificial neural networks and deep learning ○ Graph neural networks ○ Transformers and Language models	Practicals / Lab

9. Important Dates:

- **Quiz 1:** 13 March 2026 (Online)
- **Assignment**

- Release date: 16 March 2026 @23:59
- Due date: 19 April 2026 @23:59

- **Group project**
  - Release date: 16 March 2026
  - Due date: 30 April 2026 @23:59
  - Project Presentation week: 4-8 May 2026
- **Quiz 2:** 24 April 2026 (Online)
- **Test:** 25 April 2026 @08:30-10:30 & venue: Engineering Basement Section A

10. Prescribed and Recommended

Reading Prescribed Reading

- Russell, S. & Norvig, P. (2016). *Artificial intelligence: A modern approach* Global Edition. Pearson. ISBN-10: 0132071487

Recommended Reading

- Marr, B. & Ward, M. (2019). *Artificial intelligence in practice: How 50 successful companies used artificial intelligence to solve problems* (1st ed.). Chichester, West Sussex: Wiley. ISBN-10: 1119548217.

11. Technology and Equipment Readiness

- Clearly state the reliance on existing NUST equipment, ensuring no additional resources are needed.
- Include specifics on equipment that may be critical for course delivery (e.g., labs, software, or tools).

12. Student Commitments and Contact Times

- **Class Attendance:**
  - Students are expected to attend at least 95% of all face-to-face classes and practical labs, and class attendance will be taken during the class.
  - If a class is missed, it is the student's responsibility to catch up on the material covered.

- **Self-Directed Study:**
  - After each class and before the subsequent session, students should review, organize, and annotate lecture notes.
  - Further reading should be done to clarify any misunderstood topics.
  - Students must dedicate their notational hours to revision, practice, and completing assigned homework.
  - The reading list, both prescribed and recommended, should be taken seriously. Students should aim to understand prescribed material thoroughly, to the extent of being able to summaries its contents in a one-page essay without referring to the material.

- **Communications:**
  - Students must maintain active communication with their lecturers via established platforms such as MS Teams and email.
  - Course-related issues should be addressed promptly; the end of the semester is not the ideal time to resolve lingering concerns.
  - Efficient use of communication tools ensures the timely handling of all matters related to the course.

13. Lecturer Information

- Name: Mr. Naftali N Indongo
- Email: nindongo@nust.na
- Office Phone: 0612072459
- Office No: 7/015/002, IT House
- Office Hours: 08H00 am – 16H30 pm
- Consultation: By appointment via email.

Effective Start Date: Monday, 9 February 2026

Authorisation:

This course is authorised for use by:

_____ _____

Head of Department Date

ACKNOWLEDGEMENT BY STUDENT

(To be completed by all students on the course, detached from the course outline and kept on record in the department)

I, _____, student Number _____, hereby acknowledge that I have received this course outline for Artificial Intelligence (ARI711S) in semester 5 (2026) and that I have familiarized myself with its content, particularly the statement about academic honesty and integrity. I agree to abide by the policies and arrangements in this course outline.

_____ _____

Signature of student Date